Physics II — All Key Definitions

Phy 4143 · Lecture 1–4 · Quick-reference, 2-page version

Basic Quantities

Voltage: Electrical potential difference between two points in a circuit; the force that drives current. Measured in volts (V).

Current: The flow of electric charge (electrons) through a conductor. Measured in amperes (A).

Resistance: How much a material blocks the flow of current. Measured in ohms.

Ohm's Law: V = I x R — voltage is directly proportional to current and inversely proportional to resistance.

Circuit Fundamentals & Laws

Electric Circuit: A closed path for transmitting current through electrical and magnetic fields.

KCL (Kirchhoff's Current Law): The algebraic sum of currents at any node/junction is zero — current in = current out.

KVL (Kirchhoff's Voltage Law): The algebraic sum of voltages around any closed loop is zero.

Node, Junction, Branch, Loop, Mesh

Node: A point where two or more circuit elements are joined.

Junction: A point where three or more circuit elements meet; current divides here — this is where KCL applies.

Branch: The part of a network that lies between two junction points.

Loop: Any closed path in a network.

Mesh: The smallest loop that cannot be divided further.

Sources

Source: A device that converts mechanical, chemical or thermal energy into electrical energy; an active network element.

Voltage Source: A two-terminal device whose voltage is constant and independent of the current drawn from it (ideal: zero internal resistance).

Current Source: A two-terminal device that supplies the same current regardless of load voltage (ideal: infinite internal resistance).

Energy-Storing Elements

Inductor: A coil that stores energy in a magnetic field when current flows through it. v = L x (di/dt). Measured in henries (H).

Capacitor: Two conductive plates that store energy in an electric field. i = C x (dv/dt). Measured in farads (F).

Resistor Connections & Dividers

Series Connection: Resistors are in series when the same current flows through all of them. Req = R1 + R2 + ...

Parallel Connection: Resistors are in parallel when they share the same two nodes, so the same voltage appears across all of them.

Voltage Divider Rule: Splits total voltage among series resistors, proportional to resistance. Rx = resistor you want the voltage across, Rs = total series resistance.

$$V_x = \frac{R_x}{R_s} \times V$$

Current Divider Rule: Splits total current among parallel resistors. Rp = equivalent parallel resistance of the circuit, Rx = branch you want the current for.

$$I_x = \frac{R_p}{R_x} \times I$$

Transformations

Source Transformation: A practical voltage source (V, series R) can be replaced by an equivalent current source (I=V/R, R in parallel), and vice versa.

Delta to Y Transformation: Converts a Delta (triangle) resistor network into an equivalent Y (star) network.

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$$

$$R_2 = \frac{R_c R_a}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c}$$

Y to Delta Transformation: Converts a Y (star) resistor network into an equivalent Delta (triangle) network.

$$R_a = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1}$$

$$R_b = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_2}$$

$$R_c = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_3}$$